

Refined Global Fit of KamLAND Data and the Daya Bay Antineutrino Energy Spectrum

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Abstract

Recently, the JUNO experiment published its measurement of the solar neutrino oscillation parameters Δm_{21}^2 and $\sin^2 \theta_{12}$ based on 59 days of data, with central values differing by 0.2σ from those released by the KamLAND experiment in 2013. Meanwhile, short-baseline reactor neutrino oscillation experiments such as Daya Bay, RENO, and Double Chooz have observed significant deviations between the measured antineutrino spectrum and the Huber-Müller model around 5 MeV. To further investigate the impact of these deviations on the measurement of neutrino oscillation parameters, a global analysis framework is constructed that is weakly dependent on the reactor antineutrino flux model. This framework is based on the independently measured ^{235}U and ^{239}Pu fission antineutrino spectra from the Daya Bay experiment, combined with the public data from KamLAND. First, using the Huber-Müller model, the KamLAND 2013 results are successfully reproduced to within 0.1σ . Then, replacing the Huber-Müller model with the Daya Bay measured antineutrino spectra in a combined analysis, it is found that the best-fit value of the mass-squared difference Δm_{21}^2 decreases from $7.53_{-0.16}^{+0.17} \times 10^{-5} \text{ eV}^2$ to $7.50_{-0.18}^{+0.19} \times 10^{-5} \text{ eV}^2$, while the best-fit value of the mixing angle $\tan^2 \theta_{12}$ also shows a decreasing trend. This result is in better agreement with the latest JUNO measurement, suggesting that differences in the predicted reactor antineutrino spectra may be an important cause of the tension between the two experiments.

Keywords: Neutrino oscillation parameters; Reactor antineutrino spectrum; Global analysis

1. Introduction

The great success of neutrino oscillation experiments and theories in explaining the solar and atmospheric neutrino anomalies provides strong evidence that neutrinos have non-zero mass. The solar neutrino experiment SNO measured the mass-squared difference $\Delta m_{21}^2 = 5.6_{-1.4}^{+1.9} \times 10^{-5} \text{ eV}^2$ and mixing angle $\tan^2 \theta_{12} = 0.427_{-0.029}^{+0.030}$ [1]. In contrast, the long-baseline reactor neutrino oscillation experiment KamLAND reported $\Delta m_{21}^2 = 7.54_{-0.18}^{+0.19} \times 10^{-5} \text{ eV}^2$ and $\tan^2 \theta_{12} = 0.481_{-0.080}^{+0.092}$ [2]. With the constraint on θ_{13} , the KamLAND results become $\Delta m_{21}^2 = 7.54_{-0.18}^{+0.18} \times 10^{-5} \text{ eV}^2$ and $\tan^2 \theta_{12} = 0.460_{-0.056}^{+0.063}$ (from Fig. 4 in [2]). Hence, there is some tension between KamLAND and SNO in both Δm_{21}^2 and $\tan^2 \theta_{12}$, especially for Δm_{21}^2 . Very recently, the JUNO experiment published the most precise measurement of solar neutrino oscillation parameters to date, with $\Delta m_{21}^2 = 7.50_{-0.12}^{+0.12} \times 10^{-5} \text{ eV}^2$ and $\tan^2 \theta_{12} = 0.4476_{-0.0182}^{+0.0182}$ [3]. This differs from the KamLAND result by 0.2σ , indicating a mild tension.

On the other hand, short-baseline reactor neutrino oscillation experiments have revealed differences between the measured reactor antineutrino spectrum and the Huber-Müller model prediction [4, 5], both in shape and absolute flux. These differences manifest as an approximately 6% deficit in the total reactor neutrino flux and the so-called “5 MeV bump” [6–10]. Given the central role of the reactor antineutrino spectrum in reactor neutrino oscillation analyses, it is of great interest to understand how the spectral anomaly affects the fitted neutrino mass and mixing parameters. A global analysis reported

$\Delta m_{21}^2 = 7.34_{-0.16}^{+0.17} \times 10^{-5} \text{ eV}^2$ and $\tan^2 \theta_{12} = 0.437_{-0.026}^{+0.029}$ [11], where incorporating the bump feature lowered Δm_{21}^2 by about 1σ . That study used the overall energy-dependent ratio $f(E)$ between data [8] and the Huber-Müller expectation, and subsequent work adopted the Daya Bay unfolded antineutrino spectrum [10], but without a quantitative discussion of its impact on solar neutrino parameters [12]. To more accurately assess how the reactor neutrino flux expectation influences the KamLAND fit, we construct in this paper a global analysis framework that is weakly dependent on the reactor neutrino flux model.

In this study, the latest KamLAND analysis results are first reproduced. Then, by treating the ^{235}U and ^{239}Pu antineutrino spectra as free parameters and imposing constraints from the Daya Bay experiment, a refined combined fit of KamLAND data and Daya Bay antineutrino spectra is performed. For completeness, a simplified spectral correction method similar to that in Ref. [11] is also investigated. The final results show that in the combined fit, Δm_{21}^2 decreases by about 0.17σ , which is slightly smaller than the 0.29σ decrease obtained with the simplified correction. The global analysis framework developed here will be valuable for understanding the mild tension between JUNO and KamLAND, as well as for future applications of reactor antineutrino spectra, such as the unfolded spectrum from JUNO-TAO [13, 14].

2. Reproducing the KamLAND Results

The expected number of reactor neutrino events is determined by five key ingredients: the expected antineutrino spectrum $\Phi_j(E_\nu)$, the antineutrino survival probability $P_j(E_\nu, \Delta m_{21}^2, \theta_{12}, \theta_{13})$,

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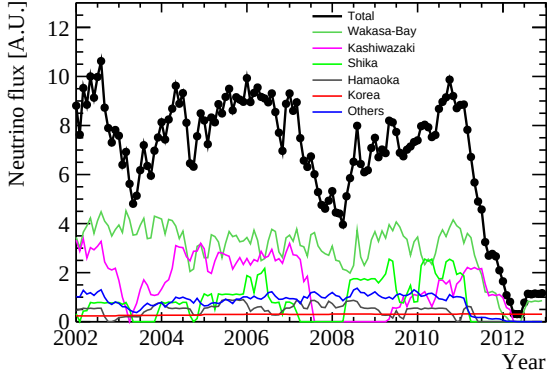


Figure 1: Reactor antineutrino flux at the KamLAND site calculated from the electricity generation data of Japanese and South Korean nuclear power plants. The total flux and contributions from different reactors are shown, in good agreement with the KamLAND results [2].

the inverse beta decay (IBD) cross section $\sigma(E_\nu)$ [15], the detector response $R(E_e, E'_e, \epsilon^E)$, and a normalization factor f . The expected events in each energy bin is given by:

$$N_i^{\text{th}}(\Delta m_{21}^2, \theta_{12}, \theta_{13}) = f \int dE_\nu \sigma(E_\nu) \sum_j \Phi_j(E_\nu) \times$$

$$P_j(E_\nu, \Delta m_{21}^2, \theta_{12}, \theta_{13}) \int dE_e R(E_e, E'_e, \epsilon^E).$$

$$\Phi_j(E_\nu) = \frac{1}{4\pi L_j^2} \sum_k \frac{f_k \cdot W_j^{\text{th}}}{\sum_l f_l \cdot e_l} \frac{d\phi_k(E_\nu)}{dE_\nu}.$$

$$R(E_e, E'_e, \epsilon^E) = \frac{1}{\sqrt{2\pi}\sigma_{E'_e}(1 + \epsilon^E)} \exp\left(-\frac{(E_e - E'_e(1 + \epsilon^E))^2}{2\sigma_{E'_e}^2(1 + \epsilon^E)}\right).$$

$$P_j(E_\nu, \Delta m_{21}^2, \theta_{12}, \theta_{13}) = \cos^4 \theta_{13} \left(1 - \sin^2(2\theta_{12M}) \sin^2 \frac{\Delta m_{21M}^2 L_j}{4E_\nu}\right) + \sin^4 \theta_{13}. \quad (1)$$

Here $d\phi_i(E_\nu)/dE_\nu$ is the antineutrino spectrum, W_j^{th} the thermal power of reactor j , L_j the baseline length, and f_k the fission fraction of isotope k in reactor j . The Huber-Müller model is used to compute $d\phi_k(E_\nu)/dE_\nu$. The detector response is a function of the prompt signal energy E_e , the true positron energy E'_e , and the linear energy scale offset ϵ^E . The quantities θ_{12M} and m_{21M}^2 are the effective mixing angle and mass-squared difference in matter, incorporating the Mikheyev-Smirnov-Wolfenstein (MSW) effect [16, 17].

The key to computing the expected event rate is the neutrino flux. Two websites [18] record the monthly electricity generation of Japanese nuclear power plants and the annual generation of South Korean plants. Since the electricity output is approximately proportional to the thermal power of a reactor, these data are used to estimate the relative neutrino flux from the 22 reactors in Japan and South Korea. As an important cross-check, Fig. 1 shows our calculated flux, which agrees well with the KamLAND result [2].

Next, we acquire the experimental data. In the KamLAND 2013 analysis, data were divided into three running periods

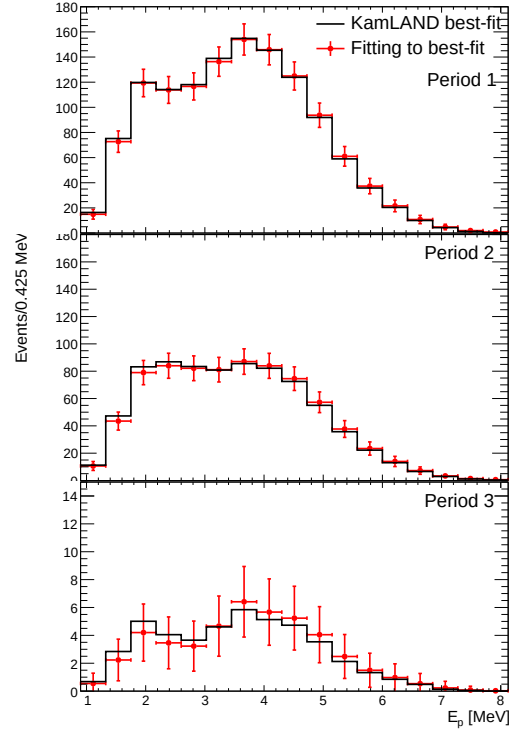


Figure 2: Fit results for the best-fit oscillation spectrum. The oscillated spectral shape and the expected no-oscillation events are in good agreement with the official results.

for a combined analysis [2]. Without neutrino oscillation, the expected number of reactor antineutrino events is 3564. The observed events, experimental errors, best-fit background values, and best-fit geo-neutrino values are given in the reference. The average fission fractions, background uncertainties, and rate systematics are also provided.

Because the official reference does not provide the expected no-oscillation events for each period, an attempt is made to recover these numbers by performing a concise χ^2 fit to the published best-fit oscillation signal with $\Delta m_{21}^2 = 7.54 \times 10^{-5} \text{ eV}^2$, $\tan^2 \theta_{12} = 0.481$ and $\sin^2 \theta_{13} = 0.01$. The χ^2 is constructed as

$$\chi^2 = \sum_{k=1}^3 \sum_{i=1}^{17} \frac{(N_{k,i}^{\text{th}}(f_k) - N_{k,i}^{\text{best}})^2}{N_{k,i}^{\text{th}}}. \quad (2)$$

In the equation, k indexes the three KamLAND data periods (P1, P2, P3). The term $N_{k,i}^{\text{th}}$ represents the expected events predicted by the model for period k in energy bin i , while $N_{k,i}^{\text{best}}$ is the corresponding observed events from the reference. The fit results are shown in Fig. 2. The expected no-oscillation events for the three periods are obtained as 2112.31, 1374.24, and 76.66, respectively. Without imposing period-to-period constraints, the total is very close to the official value. However, the fit for the third period is relatively poor, with an absolute deviation of less than one event but a relative deviation of about 20%. Given that the third period accounts for only 2% of the total statistics, this period is discarded to avoid artificially amplifying the bias due to rough handling.

Table 1: Relevant systematic uncertainties for KamLAND data (in %)

Phase	Uncertainty source				
	σ^{syst}	σ^E	σ^{Li}	σ^{anA}	σ^{anB}
P1	3.0	2.0	8.0	10.0	20.0
P2	3.5	2.0	10.0	23.0	30.0
P3	3.5	2.0	20.0	43.0	45.0

This work constructs a χ^2 function as shown in Eq. (2) based on a likelihood analysis to fit the solar neutrino oscillation parameters. The observables are the events in 17 equal prompt energy bins from 0.900 to 8.125 MeV.

$$\begin{aligned}
\chi^2 = & \sum_{k=1}^2 \sum_{i=1}^{17} \left(\frac{N_{i,k}^{\text{obs}} - N_{i,k}^{\text{th}} (1 + c_k^{\text{syst}} \epsilon^{\text{syst}} + \epsilon_i^{\text{shape}}) - N_{i,k}^{\text{B}'} - N_{i,k}^{\text{G}'}}{\sqrt{N_{i,k}^{\text{obs}}}} \right)^2 \\
& + \left(\frac{\epsilon^{\text{syst}}}{\sigma^{\text{syst}}} \right)^2 + \left(\frac{\epsilon^E}{\sigma^E} \right)^2 + \left(\frac{\epsilon^{\text{Li}}}{\sigma^{\text{Li}}} \right)^2 + \left(\frac{\epsilon^{\text{anA}}}{\sigma^{\text{anA}}} \right)^2 + \left(\frac{\epsilon^{\text{anB}}}{\sigma^{\text{anB}}} \right)^2 \\
& + \sum_{i=1, j=1}^{17} \epsilon_i^{\text{shape}} H_{ij}^{-1} \epsilon_j^{\text{shape}}, \\
N_{i,k}^{\text{B}'} = & N_{i,k}^{\text{Li}} (1 + c_k^{\text{Li}} \epsilon^{\text{Li}}) + N_{i,k}^{\text{anA}} (1 + c_k^{\text{anA}} \epsilon^{\text{anA}}) \\
& + N_{i,k}^{\text{anB}} (1 + c_k^{\text{anB}} \epsilon^{\text{anB}}) + N_{i,k}^{\text{Acc}}, \\
N_{i,k}^{\text{G}'} = & N_{i,k}^{\text{Geo}} (1 + \epsilon^{\text{Geo}}),
\end{aligned} \tag{3}$$

where $N_{i,k}^{\text{obs}}$ is the observed events in energy bin i for period k , $N_{i,k}^{\text{th}}$ is the expected reactor neutrino events, $N_{i,k}^{\text{B}'}$ is the expected events after correcting for main backgrounds, including ${}^9\text{Li}/{}^8\text{He}$, ${}^{13}\text{C}(\alpha, n){}^{16}\text{O}_{\text{g.s.}}$, (αnA) , ${}^{13}\text{C}(\alpha, n){}^{16}\text{O}^*$ (αnB), and accidental coincidences, and $N_{i,k}^{\text{G}'}$ is the geo-neutrino background contribution. $N_{i,k}^{\text{Li}}$, $N_{i,k}^{\text{anA}}$, $N_{i,k}^{\text{anB}}$, and $N_{i,k}^{\text{Acc}}$ are the expected events for each background component. σ^{syst} is the overall systematic uncertainty, σ^E is the energy scale uncertainty, and σ^{Li} , σ^{anA} , σ^{anB} are the background uncertainties. The corresponding pull parameters are ϵ^{syst} , ϵ^E , ϵ^{Li} , ϵ^{anA} , and ϵ^{anB} , respectively. The bin-to-bin spectral shape uncertainties are constrained by the covariance matrix H_{ij} with pull parameters $\epsilon_i^{\text{shape}}$. The values of uncertainties are listed in Table 1. The correlation factors c_k^{syst} , c_k^{Li} , c_k^{anA} , c_k^{anB} are defined as $\sigma_k^{\text{syst}}/\sigma_1^{\text{syst}}$, $\sigma_k^{\text{Li}}/\sigma_1^{\text{Li}}$, $\sigma_k^{\text{anA}}/\sigma_1^{\text{anA}}$, $\sigma_k^{\text{anB}}/\sigma_1^{\text{anB}}$, respectively, reflecting the correlation of the uncertainties between periods.

The quadratic terms in the equation are penalty terms for systematic error parameters, and $\sum_{i,j} \epsilon_i^{\text{shape}} H_{ij}^{-1} \epsilon_j^{\text{shape}}$ constrains the shape parameters. Penalty terms are introduced to constrain the energy scale, backgrounds, and bin-to-bin shape variations. Following KamLAND's treatment, we assume that systematic uncertainties are fully correlated between the two periods, and the geo-neutrino background is allowed to vary freely. The KamLAND detector-related uncertainty parameters used in this study are listed in Table 1, all taken from the official publication.

The reproduced allowed regions in the Δm_{21}^2 - $\tan^2 \theta_{12}$ plane are shown in Fig. 3, and the best-fit values are given in Tab. 2.

Table 2: Best-fit solar neutrino oscillation parameters from KamLAND official results and this reproduction, with θ_{13} free or constrained.

	Δm_{21}^2	$\tan^2 \theta_{12}$	$\sin^2 \theta_{13}$
	[10^{-5} eV^2]		
θ_{13} free			
Reproduced	$7.54^{+0.17}_{-0.17}$	$0.495^{+0.106}_{-0.095}$	$0.011^{+0.033}_{-0.035}$
θ_{13} free			
Official	$7.54^{+0.19}_{-0.18}$	$0.481^{+0.092}_{-0.080}$	$0.010^{+0.033}_{-0.034}$
θ_{13} constrained			
Reproduced	$7.53^{+0.17}_{-0.16}$	$0.465^{+0.067}_{-0.055}$	$0.023^{+0.002}_{-0.002}$
θ_{13} constrained			
Official	$7.53^{+0.18}_{-0.18}$	$0.460^{+0.063}_{-0.056}$	$0.023^{+0.002}_{-0.002}$

The official KamLAND results are also shown for comparison. With θ_{13} free, the reproduced Δm_{21}^2 and $\sin^2 \theta_{13}$ agree with the official results within 0.15σ , while $\tan^2 \theta_{12}$ is slightly larger. When $\sin^2 \theta_{13}$ is constrained to $0.023^{+0.002}_{-0.002}$, the reproduction becomes even more accurate, agreeing with the official results within 0.1σ . This lays a solid foundation for the subsequent combined analysis with the Daya Bay antineutrino spectra.

3. Combined Analysis

In 2019, the Daya Bay experiment published measurements of the independent antineutrino spectra of ${}^{235}\text{U}$ and ${}^{239}\text{Pu}$ [9]. They found that spectral deviations from theory appear at different confidence levels for the two isotopes. This provides the motivation and input for the combined analysis of KamLAND data and Daya Bay antineutrino spectra.

In this combined analysis, the ${}^{235}\text{U}$ and ${}^{239}\text{Pu}$ spectra are treated as free parameters, uniformly binned into 25 neutrino energy bins from 2 to 8 MeV. This introduces 50 new free parameters, which are constrained by the unfolded reactor antineutrino spectra from Daya Bay. The ${}^{238}\text{U}$ and ${}^{241}\text{Pu}$ spectra are based on the Huber-Müller model, with a bin-to-bin uncertainty of 10% assigned in the KamLAND χ^2 term. The expected no-oscillation events are kept unchanged. The Daya Bay inputs include the unfolded spectra of ${}^{235}\text{U}$ and ${}^{239}\text{Pu}$, their covariance matrix [9], and the Daya Bay detector model that converts prompt energy to deposited energy [19]. The χ^2 function

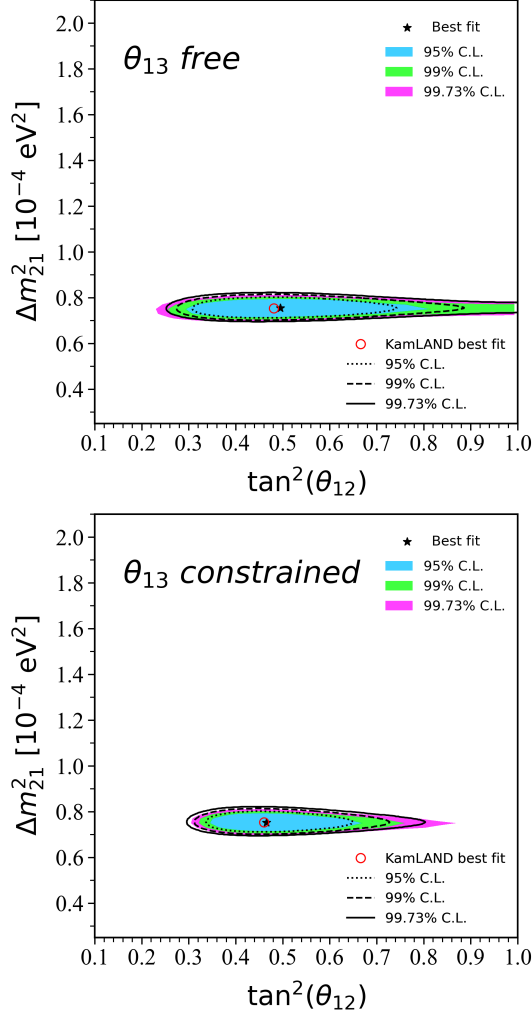


Figure 3: Allowed regions at 95%, 99%, and 99.73% confidence levels in the $(\tan^2 \theta_{12}, \Delta m^2_{21})$ plane. The filled colored areas are the reproduction results of this work, and the contours are the official results.

for the combined analysis is constructed as:

$$\chi^2 = \sum_k \sum_i \left(N_{i,k}^{\text{obs}} - N_{i,k}^{\text{th}_{235\text{U}, 239\text{Pu}}} (1 + c_k^{\text{syst}} \epsilon^{\text{syst}}) - N_{i,k}^{\text{th}_{238\text{U}, 241\text{Pu}}} (1 + c_k^{\text{syst}} \epsilon^{\text{syst}} + \epsilon_i^{\text{shape}}) - N_{i,k}^{\text{B}'} - N_{i,k}^{\text{G}'} \right)^2 / N_{i,k}^{\text{obs}} + \left(\frac{\epsilon^{\text{syst}}}{\sigma^{\text{syst}}} \right)^2 + \left(\frac{\epsilon^E}{\sigma^E} \right)^2 + \left(\frac{\epsilon^{\text{Li}}}{\sigma^{\text{Li}}} \right)^2 + \left(\frac{\epsilon^{\alpha\text{N}A}}{\sigma^{\alpha\text{N}A}} \right)^2 + \left(\frac{\epsilon^{\alpha\text{N}B}}{\sigma^{\alpha\text{N}B}} \right)^2 + \sum_i \left(\frac{\epsilon_i^{\text{shape}}}{\sigma^{\text{shape}}} \right)^2 + \sum_{r,s} \left(\frac{N_r^{\text{th}}}{N_r^{\text{extr}}} - 1 \right) B_{rs}^{-1} \left(\frac{N_s^{\text{th}}}{N_s^{\text{extr}}} - 1 \right). \quad (4)$$

where:

- $N_{i,k}^{\text{th}_{235\text{U}, 239\text{Pu}}}$: The expected events in energy bin i of period k from ^{235}U and ^{239}Pu fission. The spectral shape of this term is constrained by the Daya Bay unfolded reactor antineutrino spectra.
- $N_{i,k}^{\text{th}_{238\text{U}, 241\text{Pu}}}$: The theoretical expected events in energy bin

i of period k from ^{238}U and ^{241}Pu fission.

- σ^{shape} : The normalization standard deviation of the spectral shape uncertainty for ^{238}U and ^{241}Pu . Following the Huber-Müller model, a conservative value of 10% is used.
- N_r^{th} : The expected events in the r -th prompt energy bin of the Daya Bay experiment. Indices r and s run over the 52 bins of the unfolded antineutrino spectra from Daya Bay (26 bins for each of ^{235}U and ^{239}Pu).
- N_r^{extr} : The prompt energy spectrum of ^{235}U and ^{239}Pu from the Daya Bay experiment.
- B_{rs} : The covariance matrix of the unfolded ^{235}U and ^{239}Pu spectra.

The last term in the above expression contains the measured prompt energy spectra of ^{235}U and ^{239}Pu from Daya Bay, which, in the joint fit, constrains the variations of the two isotope spectra and thereby propagates the impact to the oscillation parameters. In this combined analysis, the link between the KamLAND data and the Daya Bay antineutrino spectra is the 50 free spectral parameters.

In addition, a simplified combined analysis is performed that directly uses the unfolded antineutrino spectrum from Daya Bay to correct the KamLAND expectation via the ratio of data to theoretical prediction. This simplification is justified by the fact that the average fission fractions of the two experiments are quite close. For simplicity and conservativeness, the covariance matrix and shape uncertainties of the Huber-Müller model are kept unchanged in this simplified analysis.

The results of the combined fits are summarized in Figs. 4, 5 and Tab. 3. Both combined analysis schemes (with different χ^2 formulations and reactor antineutrino inputs) show a consistent trend in modifying the oscillation parameters: introducing the Daya Bay measured antineutrino spectra systematically lowers the best-fit central values of Δm^2_{21} and $\tan^2 \theta_{12}$.

The results of the full combined fit using the 2019 Daya Bay independent spectra (+DayaBay19) are first examined. With θ_{13} free, compared to the pure KamLAND reproduction, Δm^2_{21} decreases from $7.54^{+0.17}_{-0.17} \times 10^{-5} \text{ eV}^2$ to $7.49^{+0.19}_{-0.19} \times 10^{-5} \text{ eV}^2$, a shift of approximately 0.26σ relative to the largest lower uncertainty. $\tan^2 \theta_{12}$ drops significantly from $0.495^{+0.106}_{-0.095}$ to $0.416^{+0.103}_{-0.082}$, a shift of about 0.83σ . Meanwhile, $\sin^2 \theta_{13}$ increases from $0.011^{+0.033}_{-0.035}$ to $0.034^{+0.033}_{-0.034}$, a change of nearly 1σ . However, given KamLAND's very low sensitivity to θ_{13} , the fitted θ_{13} has large uncertainties and limited physical significance. In the constrained θ_{13} mode, Δm^2_{21} decreases from $7.53^{+0.17}_{-0.16} \times 10^{-5} \text{ eV}^2$ to $7.50^{+0.19}_{-0.18} \times 10^{-5} \text{ eV}^2$ (a shift of about 0.17σ), and $\tan^2 \theta_{12}$ from $0.466^{+0.067}_{-0.055}$ to $0.439^{+0.064}_{-0.053}$ (a shift of about 0.49σ).

Comparing the simplified combined method (+DayaBay17) with the full combined method (+DayaBay19), the trend in oscillation parameters is consistent. In both the free and constrained θ_{13} cases, the central value of Δm^2_{21} from the full fit is slightly higher than that from the simplified fit, with a difference of about $0.01 \times 10^{-5} \text{ eV}^2$ ($\sim 0.06\sigma$). The central value of $\tan^2 \theta_{12}$ from the full fit is slightly lower than that from the simplified fit, with differences of about 0.09σ and 0.15σ for free

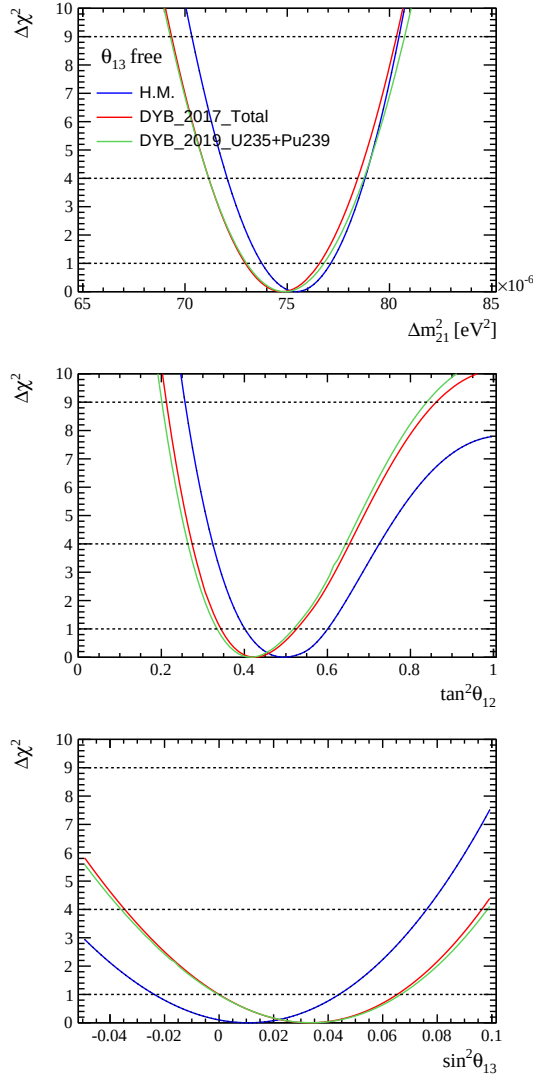


Figure 4: Projections of $\Delta\chi^2$ onto each oscillation parameter in the free θ_{13} case. After including the Daya Bay measured antineutrino spectra, the best-fit central values of Δm_{21}^2 and $\tan^2 \theta_{12}$ decrease systematically.

and constrained θ_{13} cases, respectively. Moreover, the uncertainties in the full fit are slightly larger than those in the simplified fit. This is because the 2019 Daya Bay measurement provides isotope-decomposed spectra and a full covariance matrix, which introduces more independent systematic uncertainties and statistical degrees of freedom compared to the global spectral ratio constraint used in 2017. Consequently, while the central values of the oscillation parameters are suppressed, their uncertainties are moderately enlarged.

4. Conclusion

Based on the KamLAND 2013 long-baseline reactor neutrino oscillation data and the Daya Bay 2019 short-baseline antineutrino spectra of ^{235}U and ^{239}Pu , a detailed combined analysis framework has been constructed that incorporates spectral shape constraints and systematic error correlations. This frame-

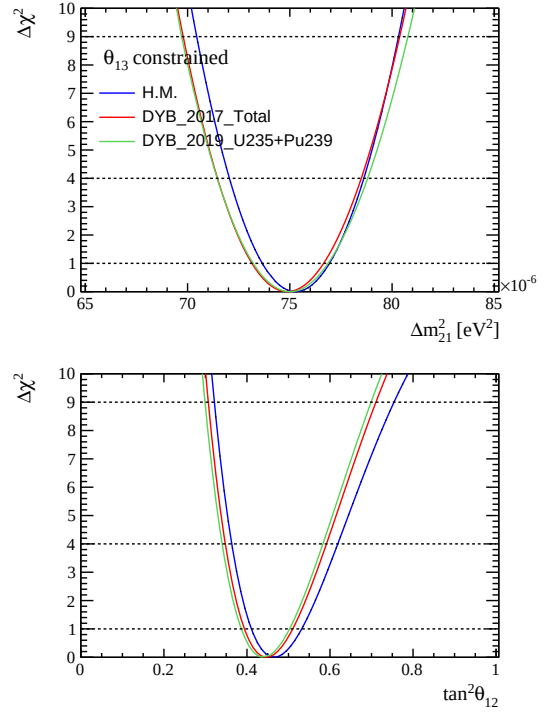


Figure 5: Projections of $\Delta\chi^2$ onto each oscillation parameter in the constrained θ_{13} case. After including the Daya Bay measured antineutrino spectra, the best-fit central values of Δm_{21}^2 and $\tan^2 \theta_{12}$ decrease systematically.

work systematically investigates the impact of reactor antineutrino spectrum anomalies on the neutrino oscillation parameters Δm_{21}^2 and $\tan^2 \theta_{12}$. In the constrained θ_{13} mode, the best-fit solar oscillation parameters are $\Delta m_{21}^2 = 7.50^{+0.19}_{-0.18} \times 10^{-5} \text{ eV}^2$ and $\tan^2 \theta_{12} = 0.439^{+0.064}_{-0.053}$. Compared to the KamLAND standalone fit using the Huber-Müller theoretical spectrum, the central values of Δm_{21}^2 and $\tan^2 \theta_{12}$ decrease by about 0.2σ and 0.4σ , respectively, bringing the measurements into better agreement with the latest high-precision results from JUNO.

This study demonstrates that measurements of the reactor antineutrino spectrum can effectively break the dependence of reactor neutrino oscillation experiments on spectral models and partially alleviate the tension between KamLAND, JUNO and SNO. Furthermore, the combined KamLAND–Daya Bay analysis model developed here can serve as a reference for spectral model corrections in reactor neutrino experiments such as JUNO.

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Table 3: Best-fit results of the oscillation parameters Δm_{21}^2 , $\tan^2 \theta_{12}$, and $\sin^2 \theta_{13}$ obtained from the KamLAND-only reproduction and from the combined analyses with the Daya Bay 2017 (simplified) and 2019 (full) antineutrino spectra, under both free and constrained θ_{13} scenarios.

	θ_{13} free			θ_{13} constrained		
	Reproduction	+DayaBay(17)	+DayaBay(19)	Reproduction	+DayaBay(17)	+DayaBay(19)
Δm_{21}^2 [10^{-5} eV 2]	$7.54^{+0.17}_{-0.17}$	$7.48^{+0.18}_{-0.18}$	$7.49^{+0.19}_{-0.19}$	$7.53^{+0.17}_{-0.16}$	$7.49^{+0.18}_{-0.17}$	$7.50^{+0.19}_{-0.18}$
$\tan^2 \theta_{12}$	$0.495^{+0.106}_{-0.095}$	$0.425^{+0.102}_{-0.081}$	$0.416^{+0.103}_{-0.082}$	$0.466^{+0.067}_{-0.055}$	$0.447^{+0.064}_{-0.053}$	$0.439^{+0.064}_{-0.053}$
$\sin^2 \theta_{13}$	$0.011^{+0.033}_{-0.035}$	$0.033^{+0.032}_{-0.033}$	$0.034^{+0.033}_{-0.034}$	$0.023^{+0.002}_{-0.002}$	$0.023^{+0.002}_{-0.002}$	$0.023^{+0.002}_{-0.002}$

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